NEWTON SCHOOL OF TECHNOLOGY

Data Visualization & Analytics — Capstone Project 2

Road Accident Analysis

Identifying Patterns, Risk Factors, and Evidence-Based Interventions

Sector	Public Safety & Transportation
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1. Executive Summary

Problem

Traffic accidents in Addis Ababa, Ethiopia impose enormous human and economic costs. Without granular analysis of when, where, and why accidents happen, public safety resources cannot be deployed effectively. This project addresses that gap by transforming raw police incident records into a structured, decision-ready evidence base.

Approach

A structured Python-to-Tableau pipeline was built — raw data cleaned with Pandas across Notebooks 01–02, analysed in Notebooks 03–04, and surfaced through four interactive Tableau Public dashboards covering severity, driver profiles, risk factors, and temporal patterns.

Key Findings

- 84.59% of accidents result in slight injuries, 14.13% serious, and 1.29% fatal — the 15.41% serious-and-fatal tail carries the highest societal cost.
- Accident frequency peaks at 17:00 (1,224 cases), with Friday the highest-volume day of the week.
- Drivers aged 18–30 account for 47.2% of incidents; the 2–5 year experience band shows the highest fatal rate at 1.76% — an overconfidence zone.
- Weekend accidents are 2.45x more fatal than weekday accidents; night-time driving produces a severity rate of 20.18%.
- 84% of accidents occur under normal weather — confirming this is a behaviour-driven problem, not an environment-driven one.

Key Recommendations

- Concentrate enforcement on Friday evenings and weekend nights to target the 2.45x fatality multiplier.
- Introduce graduated licensing for drivers with 0–5 years of experience to address the overconfidence zone.
- Deploy camera-based distance monitoring — 'no proper distancing' causes 18.56% of all incidents.
- Upgrade street lighting on identified night-time hotspot corridors to reduce the 20.18% night severity rate.

2. Sector & Business Context

Sector Overview

Road traffic safety sits at the intersection of public health, urban planning, and transportation policy. The World Health Organization estimates approximately 1.19 million annual road-traffic deaths globally, with over 90% occurring in low- and middle-income countries. Urban migration is outpacing infrastructure investment, and driver behaviour remains the leading proximate cause of accidents. Most agencies lack the analytical capacity to turn raw incident records into evidence-based policy — a gap this project directly addresses.

Decision-Maker / Stakeholder

The primary decision-maker served is a senior traffic safety administrator at a municipal or national road safety authority. This stakeholder requires evidence on where risk concentrates, why severity varies across conditions, and which interventions deliver the greatest casualty reduction per unit of expenditure.

Why This Problem Matters

The business value is concrete: even a 10% reduction in serious accident rates translates into hundreds of lives saved annually and significant reductions in healthcare, productivity, and emergency service costs. The dataset covers Addis Ababa (2017–2020) — a rapidly urbanising city where the gap between growing road traffic and safety infrastructure is widening. Evidence-based prioritisation of enforcement, licensing reform, and infrastructure investment is urgently needed.

3. Problem Statement & Objectives

Formal Problem Definition

Analyse road accident data from Addis Ababa, Ethiopia (2017–2020) to identify patterns and environmental/demographic conditions driving accident severity — and produce prioritised, evidence-based recommendations to reduce fatalities and serious injuries.

Objectives

- Identify high-risk time periods and locations where accidents concentrate.
- Analyse driver risk factors — age band, driving experience, and gender — and their relationship to accident severity.
- Evaluate how lighting, weather, and road surface conditions influence accident outcomes.
- Build four interactive Tableau dashboards to surface findings to non-technical stakeholders.
- Produce five or more prioritised safety recommendations with quantified expected impact.

Scope

In scope: Data extraction, cleaning, EDA, statistical modelling, KPI framework, and dashboard design for the Addis Ababa RTA dataset.

Out of scope: Real-time data integration, GPS-level spatial mapping, and production machine learning model deployment.

Success Criteria

- Cleaned and validated dataset with documented cleaning log.
- Key statistics cross-validated across Python and Tableau dashboards.
- Ten or more decision-relevant insights documented in structured format.
- Four live Tableau dashboards with working filters and KPI panels.
- Five or more feasible recommendations with quantified expected impact estimates.

4. Data Description

Source Citation & Access

Source: Road Traffic Accidents (RTA) Dataset — Addis Ababa Sub-city Police Departments, published on Kaggle.

Access: https://www.kaggle.com/datasets/saurabhshahane/road-traffic-accidents

Dataset Size & Coverage

Raw Dataset Size	12,316 rows x 32 columns
Cleaned Dataset Size	12,295 rows x 22 columns (18 original + 4 derived)
Coverage Period	2017–2020, Addis Ababa, Ethiopia
Target Variable	Accident_severity (Slight / Serious / Fatal)
Numeric Columns	Number_of_casualties, Number_of_vehicles_involved

Key Columns

Column	Description	Data Type
Time	Time of accident (HH:MM:SS)	Datetime
Day_of_week	Day accident occurred	Categorical
Age_band_of_driver	Driver age group (e.g., 18–30, 31–50)	Ordinal
Driving_experience	Years of driving experience	Ordinal
Sex_of_driver	Driver gender	Categorical
Type_of_vehicle	Category of vehicle involved	Categorical
Types_of_Junction	Junction type at accident site	Categorical
Weather_conditions	Clear, rain, fog, etc.	Categorical
Light_conditions	Daylight, darkness w/w-o lighting	Categorical
Type_of_collision	Rear-end, head-on, rollover, etc.	Categorical
Cause_of_accident	Primary attributed cause	Categorical
Accident_severity	TARGET: Slight / Serious / Fatal	Ordinal

Data Quality Issues

- Police-record under-reporting bias — minor unreported accidents are absent, likely inflating severity proportions.
- Geographic specificity — findings reflect Addis Ababa's specific road environment; generalisability requires local validation.
- High 'Unknown' proportions in several columns reduce sub-group precision.
- No GPS coordinates — location analysis limited to area type and junction configuration descriptions.

5. Cleaning & Transformation

Major Cleaning Steps

Step	Action	Detail
Extraction	Load raw CSV; schema inspection	Row count, null counts, type verification in 01_extraction.ipynb
Missing Values	Mode imputation (<5%); indicator flag (5–20%); exclusion (>20%)	Documented in 02_cleaning.ipynb cleaning log
Outliers	IQR method on numeric columns	Mass-casualty records retained; implausible zeros removed
Standardisation	Time to 4-hour bins; strip whitespace; title-case normalisation	Severity encoded: Slight=1, Serious=2, Fatal=3
Column Removal	14 columns removed (30–40%+ nulls or low relevance)	Includes: Defect_of_vehicle, Educational_level, Service_year_of_vehicle, etc.
Feature Engineering	4 derived columns created	Hour, Time_period, severity_score, is_weekend
Load	Export cleaned CSV to data/processed/	05_final_load_prep.ipynb used for final validation and export

Derived Columns

Derived Column	Logic	Business Meaning
Hour	Time.dt.hour	Extracts hour for time-based analysis
Time_period	Night (0–6), Morning (6–12), Afternoon (12–18), Evening (18–24)	Identifies high-risk time slots
severity_score	slight=1, serious=2, fatal=3	Converts severity into numeric KPI
is_weekend	Day_of_week in (Saturday, Sunday)	Helps compare weekday vs weekend risk

Assumptions Made

- Missing time entries were excluded from Rush Hour flag computation rather than imputed.
- Missing weather entries retained as 'Unknown' category to avoid introducing bias.
- Mode imputation applied globally under the missing-at-random (MAR) assumption.
- Removed invalid merged entry: 'rural village areasoffice areas' — data corruption artifact.

Output Dataset Description

The cleaned dataset contains 12,295 rows and 22 columns (21 rows removed, 14 columns dropped, 4 derived columns added). All categorical values are standardised to lowercase with trimmed whitespace. The dataset is class-imbalanced — 84.59% of records are slight injury — which should be noted when interpreting any predictive modelling outputs.

6. KPI Framework

All KPIs were computed in Python (Notebooks 03 & 04) and cross-validated against Tableau dashboard values. The framework prioritises severity-weighted outcomes over raw accident counts, consistent with the project objective of casualty reduction.

KPI	Formula	Business Relevance	Value (Dataset)
Total Accident Count	COUNT(records)	Volume baseline for capacity planning	12,295
Fatal Accident Rate	(Fatal / Total) x 100	Primary outcome measure	1.29%
Serious Injury Rate	(Serious / Total) x 100	Severity-weighted risk assessment	14.13%
Night-time Severity Rate	(Night Serious+Fatal / Night Total) x 100	Temporal enforcement targeting	20.18%
Weekend Fatality Multiplier	Weekend Fatal Rate / Weekday Fatal Rate	Weekend enforcement ROI	2.45x
Peak Hour Accident Share	(Rush-hour accidents / Total) x 100	Enforcement scheduling	17:00 = 1,224 cases
High-Risk Driver Share	(Drivers <5 yrs exp / Total) x 100	Licensing policy targeting	47.2% (18–30 band)
Top Cause Share	(No distancing / Total) x 100	Infrastructure investment priority	18.56%

7. Exploratory Analysis

EDA was conducted in notebooks/03_EDA.ipynb, covering five analytical dimensions. All patterns were independently validated against Tableau dashboard outputs.

7.1 Temporal Patterns

Accident frequency peaks at 17:00 with 1,224 cases — the single highest hour in the dataset. Friday registers the highest weekly volume. Weekend accidents carry a 2.30% fatal rate versus 0.94% on weekdays — a 2.45x fatality multiplier — driven by leisure driving and night-time risk contexts. The temporal concentration of accidents is highly predictable, making enforcement ROI calculable and high.

7.2 Severity Distribution

84.59% slight injuries | 14.13% serious | 1.29% fatal. The combined serious-and-fatal rate of 15.41% is the intervention priority — it concentrates sharply in identifiable driver and time sub-groups rather than being uniformly distributed across the dataset.

7.3 Driver Profile Analysis

The 18–30 age band accounts for 47.2% of all accidents; male drivers represent 94.3% of all involvement. The 2–5 year experience band has the highest fatal rate at 1.76% — exceeding both novices (<1 year) and experienced drivers (>10 years), consistent with the overconfidence effect documented in road safety literature. This non-linear relationship between experience and severity is a key policy-relevant finding.

7.4 Collision Type & Cause

Vehicle-to-vehicle collisions account for 72.6% of all accidents. The single largest attributed cause is 'no proper distancing' at 18.56% of all incidents. Lorries show the highest severity rate at approximately 19.6% among all vehicle types — roughly five percentage points above the dataset average.

7.5 Environmental Conditions

Night-time severity rate is 20.18% versus the 15.41% dataset average. Critically, 84% of accidents occur under normal weather conditions — firmly establishing this as a behaviour-driven, not weather-driven, problem. This finding has significant policy implications: enforcement and driver education will outperform weather-oriented infrastructure spending on a per-rand cost-benefit basis.

8. Statistical Analysis

All statistical analysis was conducted in notebooks/04_statistical_analysis.ipynb using Python (scipy, pandas, sklearn). Methods include descriptive statistics, hypothesis testing, K-Means clustering, and Spearman correlation.

8.1 Descriptive Statistics

Casualty counts are positively skewed, with the mean exceeding the median — indicating a small number of high-casualty accidents pulling the average up. Univariate analysis covered mean, median, mode, variance, standard deviation, min, max, and range for all numeric features. IQR analysis confirmed outliers in casualty counts were genuine mass-casualty events, not data errors, and were retained.

8.2 Hypothesis Testing (Chi-Squared)

- Experience vs Severity: $p < 0.001$ — the 2–5 year band's 1.76% fatal rate is a genuine structural pattern, not sampling variation.
- Light Conditions vs Severity: $p < 0.01$ — night-time without lighting significantly elevates fatal outcomes.
- Weekend vs Weekday Fatal Rate: $p < 0.05$ — the 2.45x multiplier is statistically significant and policy-relevant.

8.3 Segmentation (K-Means Clustering)

Three clusters were identified from the accident data:

- Cluster 1 — 'Peak-Hour Urban High-Volume': Accidents at 17:00, Friday, vehicle-to-vehicle collisions. Highest frequency, moderate severity.
- Cluster 2 — 'Night-Time High-Severity': 20.18% severity rate, weekend multiplier. Highest fatality concentration.
- Cluster 3 — 'Normal-Condition Slight-Injury Baseline': 84.59% slight injury distribution. Bulk volume, lowest per-incident societal cost.

Clusters 1 and 2 drive the overwhelming majority of serious and fatal outcomes and represent the primary intervention targets.

8.4 Correlation & Behavioural Analysis

Spearman correlation confirms that driving experience negatively correlates with accident severity overall, but with a non-linear peak at 2–5 years — the overconfidence zone. Since 84% of accidents occur under normal weather, behavioural factors (following distance, experience level, time-of-day risk behaviour) explain the dominant share of variance. This means enforcement and education interventions deliver greater aggregate impact than weather-focused engineering investment.

9. Dashboard Walkthrough

Four interactive dashboards were published within the FINALDVA1 workbook on Tableau Public. All dashboards include filters for time period, area, day of week, driver attributes, and severity level, with dynamic KPI panels that update on filter selection.

Full Project Dashboard: https://public.tableau.com/app/profile/arya.pandey1785/viz/FINALDVA/DVAProject

#	Dashboard Title	Key Insight Surfaced	Tableau URL
1	Accident Severity Analysis	20.18% night severity; 84% normal-weather dominance; 18.56% distancing cause; integrated multi-filter executive view	.../Dashboard1
2	Driver Profile & Incident Risk	18–30 age heatmap; overconfidence zone at 2–5 yr experience; 2.30% vs 0.94% weekend vs weekday fatal rate	.../Dashboard2
3	Accident Risk & Severity	84.59% slight / 14.13% serious / 1.29% fatal KPIs; severity-by-collision-type and vehicle-type breakdown	.../Dashboard3
4	Time & Location Analysis	17:00 peak (1,224 cases); Friday highest-volume; Y/T junction severity concentration; hourly heatmap	.../Dashboard4

Dashboard Objective

Each dashboard serves a specific analytical audience. Dashboard 1 provides an executive overview of severity distribution. Dashboard 2 enables driver-risk profiling for licensing policy. Dashboard 3 supports infrastructure and enforcement prioritisation by risk factor. Dashboard 4 enables temporal and spatial targeting for field resource deployment.

Filters & Interactivity

All dashboards support cross-filter interactivity — clicking any chart element filters all other panels on the same dashboard. Available global filters include: Time Period (Morning / Afternoon / Evening / Night), Day of Week, Driver Age Band, Driving Experience, Accident Severity Level, and Area Type.

10. Key Insights

The following ten insights are stated in decision language — oriented toward the traffic safety administrator as the primary stakeholder. Each is directly supported by statistical evidence from the analysis.

#	Insight (Decision Language)
1	Weekend accidents are 2.45x more fatal (2.30% vs 0.94%). Weekend-specific enforcement must be resourced independently — not treated as an extension of weekday operations.
2	17:00 is the single highest-risk hour (1,224 cases); Friday is the highest-risk day. Risk is predictable — enforcement ROI is calculable and high.
3	Night-time driving has the highest severity rate at 20.18%. Night enforcement and street lighting on dark hotspot corridors are the two highest-leverage temporal interventions.
4	84% of accidents occur under normal weather. This is a behaviour-driven problem — enforcement and driver education will outperform weather-oriented engineering investment.
5	The 18–30 age band drives 47.2% of accidents; 94.3% of drivers are male. Targeted graduated licensing and hazard training for young male drivers address the highest-volume risk segment.
6	The 2–5 year experience band has the highest fatal rate (1.76%) — the overconfidence zone. Refresher assessments at the 2-year mark are directly supported by this data.
7	No proper distancing is the largest single cause (18.56%) and vehicle-to-vehicle collisions account for 72.6%. Camera-based distance enforcement is the highest-return single infrastructure intervention.
8	Lorries show the highest severity rate (~19.6%). Stricter operational monitoring and peak-hour route restrictions are warranted for heavy vehicle management.
9	Y/T-shaped junctions drive disproportionate serious outcomes. Junction redesign, signage upgrades, and speed control should be prioritised in the infrastructure pipeline.
10	The 15.41% serious-and-fatal share carries the overwhelming societal cost. Success should be measured by serious-and-fatal rate reduction — not total accident count.

11. Recommendations

Five recommendations are presented in priority order. Each is grounded in specific statistical findings and includes an expected impact range drawn from comparable road safety interventions in the literature.

Priority	Recommendation	Insight Basis	Action	Expected Impact
1 — High	Weekend & Night Enforcement	2.45x weekend fatal multiplier; 17:00 peak = 1,224 cases; 20.18% night severity	Deploy enforcement Friday 16:00–20:00 and weekend 22:00–03:00; speed monitoring at 17:00 peak	~20% fatal reduction on target corridors within 12 months
2 — High	Graduated Licensing	18–30 = 47.2% of accidents; 2–5 yr band = 1.76% fatal rate (highest)	Restrict night driving and high-speed roads for first 5 years; mandatory refresher at 2-year mark	20–40% fatal reduction in target cohort within 3 years
3 — High	Camera Distance Enforcement	No proper distancing = 18.56%; vehicle-to-vehicle = 72.6%	Deploy following-distance cameras on top five highest-frequency corridors at 17:00 peak	15–20% reduction in vehicle-to-vehicle serious collisions within 12 months
4 — High	Street Lighting Upgrades	Night severity 20.18% vs 15.41% dataset average	Solar LED lighting at identified night-time hotspot corridors	20–30% reduction in night-time serious injuries at lit sites
5 — Medium	Heavy Vehicle Regulation	Lorry severity ~19.6% — highest of all vehicle types	Mandatory electronic monitoring (speed, rest, load); restrict lorry routes during 17:00 peak	8–12% severity-weighted reduction on target corridors

12. Limitations & Next Steps

Data Limitations

- Geographic specificity: findings reflect Addis Ababa's road environment — application elsewhere requires local data validation before policy transfer.
- Under-reporting bias: police records skew toward severe incidents; slight unreported accidents are absent, potentially inflating severity proportions.
- No GPS coordinates: location analysis limited to area type and junction configuration — not specific addressable sites or road segments.
- High 'Unknown' proportions in several columns reduce sub-group precision; findings in those groups should be treated as indicative rather than definitive.

Method Limitations

- Observational design: risk ratios identify associations, not causal mechanisms. Causal interpretations are informed by domain knowledge and published road safety literature.
- Class imbalance: 84.59% slight-injury dominance may affect any predictive model trained on this dataset without resampling techniques.
- K-Means clustering assumes spherical clusters and Euclidean distance — appropriate for exploratory segmentation but not for inference.

Suggested Future Work

- GPS integration: geospatial hotspot analysis (KDE / DBSCAN) to produce address-level risk maps for infrastructure prioritisation.
- Predictive severity model: multiclass classifier (gradient-boosted trees / random forest) to predict severity from incident features for real-time emergency resource dispatch.
- Longitudinal extension: time-series analysis beyond 2017–2020 to evaluate policy intervention impact empirically.
- Real-time dashboard: live data ingestion from police reporting systems feeding an operational Tableau or Power BI dashboard for proactive resource deployment.

13. Contribution Matrix

Contributions are verifiable through GitHub Insights, pull request history, and committed file artifacts. All claims below match the GitHub commit and PR history of the repository.

Team Member	Role	Dataset & Sourcing	ETL & Cleaning	EDA & Analysis	Statistical Analysis	Tableau Dashboard	Report Writing	PPT & Viva
Suhaani Garg	Project Lead	Y	Y			Y	Y	
Krishna Dave	Strategy Lead				Y	Y		
Sathvik	ETL Lead	Y	Y			Y		
Hardik Maheshwari	Analysis Lead			Y	Y			
Anushka Tyagi	PPT & Quality Lead					Y		Y
Keshav	Data Lead			Y			Y	
Arya Pandey	Visualization Lead				Y	Y		

Declaration: We confirm that the contribution details above are accurate and verifiable through GitHub Insights, pull request history, and submitted project artifacts.

Appendix — Repository & Dashboard Reference

Repository Structure

- notebooks/01_extraction.ipynb — Data loading, schema inspection, quality report
- notebooks/02_cleaning.ipynb — Full cleaning pipeline (missing values, outliers, standardisation, feature engineering)
- notebooks/03_EDA.ipynb — EDA: distribution analyses, charts, correlation matrices
- notebooks/04_statistical_analysis.ipynb — Hypothesis tests, clustering, risk ratios
- notebooks/05_final_load_prep.ipynb — Export cleaned data to data/processed/ for Tableau
- scripts/etl_pipeline.py — Automated ETL pipeline (CLI)
- docs/data_dictionary.md — Column definitions & data quality notes

Tableau Dashboard Links

Full Project Dashboard: https://public.tableau.com/app/profile/arya.pandey1785/viz/FINALDVA/DVAProject

Dashboard 1 (Severity): https://public.tableau.com/app/profile/krishna.dave3250/viz/FINALDVA1/Dashboard1

Dashboard 2 (Driver Profile):
https://public.tableau.com/app/profile/krishna.dave3250/viz/FINALDVA1/Dashboard2

Dashboard 3 (Risk & Severity):
https://public.tableau.com/app/profile/krishna.dave3250/viz/FINALDVA1/Dashboard3

Dashboard 4 (Time & Location):
https://public.tableau.com/app/profile/krishna.dave3250/viz/FINALDVA1/Dashboard4

GitHub Repository: https://github.com/Suhaani-19/SectionA_Group7_RoadAccidents